

Manufacturing Integration Framework: A SOA Perspective on Manufacturing

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Abstract: Today's society, driven by aggressive marketing campaigns, shows an appetite for everything new and state of the art. The life span of a given product type, in these conditions, becomes shorter and shorter. To remain competitive, manufacturing enterprises need to keep up with these dynamic market variations, by always adjusting the offer according to competition. On top of this, the poor economic conditions add additional pressure on optimization and efficiency in order to cut costs and maximize profit. Enterprises have understood that the "time to market" becomes a decisive factor that separates successful enterprises from lagers. This time to market can be shortened only by increasing flexibility in the manufacturing process. The purpose of this paper is to present IBM's framework for manufacturing integration, which matches plant floor solutions with business systems and suppliers. IBM's manufacturing integration framework focuses mainly on achieving flexibility by enabling a low coupling design of the entire enterprise system through leveraging of Service Oriented Architecture and Manufacturing Service Bus as best practices.

Keywords: IBM Manufacturing Integration Framework, SOA, BPEL, ISA95

1. INTRODUCTION

The last decade we have witness a fast evolution in the enterprise applications space, driven primarily by the need to consolidate the information flow at enterprise level. Large enterprises understood that in order to remain competitive and expand in a global market, they need to adapt to the new rules of the game that the expansion of the Internet have introduced. On the other hand, advances in the middleware technologies as J2EE, .NET and RDBMS systems have offered a solid platform, and maybe the first such platform that ever existed, composed by both industry standards and software itself, that would assure long term stability at the middleware level. The conjunction of these factors created a favourable environment for emerging concepts like Service Oriented Architecture (SOA) and Enterprise Service Bus (ESB).

SOA can be seen and understood in many ways and it is a fact that practitioners have different understanding of SOA. First, at a high level, SOA represents an enterprise architecture best practice, which requires business componentization around business services. This assures a clear delimitation in terms of service provider and service consumer, across every cross section of the enterprise business processes. Then, software vendors use the term SOA to market various products that facilitate the definition of services as well as execution of business processes in terms of orchestrations and choreographies. Finally the software developer sees SOA from a technical point of view, in terms of web services, payloads, messaging, BPEL and so on (Fiammante 2010, Catts 2009).

Regardless of the various views on SOA, the industry has proven by now that it is a best practice model for building enterprise applications (Forrester Research 2005). The main

advantage that SOA brings to the table is the flexibility of the integration, or in other words, the flexibility of the enterprise application, which is assured by promoting loose coupling between the business components and the applications implementing them (Brown 2008).

A second concept that was built on SOA foundation is the ESB. Historically the ESB has evolved from three different directions. One direction was the early message oriented middleware platforms that offered mainly temporal decoupling by introducing asynchronous messaging capabilities. Another direction was the web services standard that allowed a decoupling by promoting open communication standards (SOAP, XML and so on). At the same time, system integrators have developed many application integration toolkits in order to respond to the enterprise demand for tighter integration of various systems. These toolkits consisted mainly in software packages responsible for translation and mediation of data, allowing the information flow between the systems being integrated. All these early stages of ESB have crystalized to form the commercial ESB implementations that are widely available today. Like in the case of SOA, ESB can also be perceived from an enterprise architecture perspective or from a pure technical perspective (Erl et. al 2012).

At an architectural level, ESB is providing a uniform and centralized information flow across all business components. This allows optimization of business units that are internal for the enterprise by allowing the creation of a utility layer. At the same time it enables a federated dataflow across partner organizations (Keen et. al 2004).

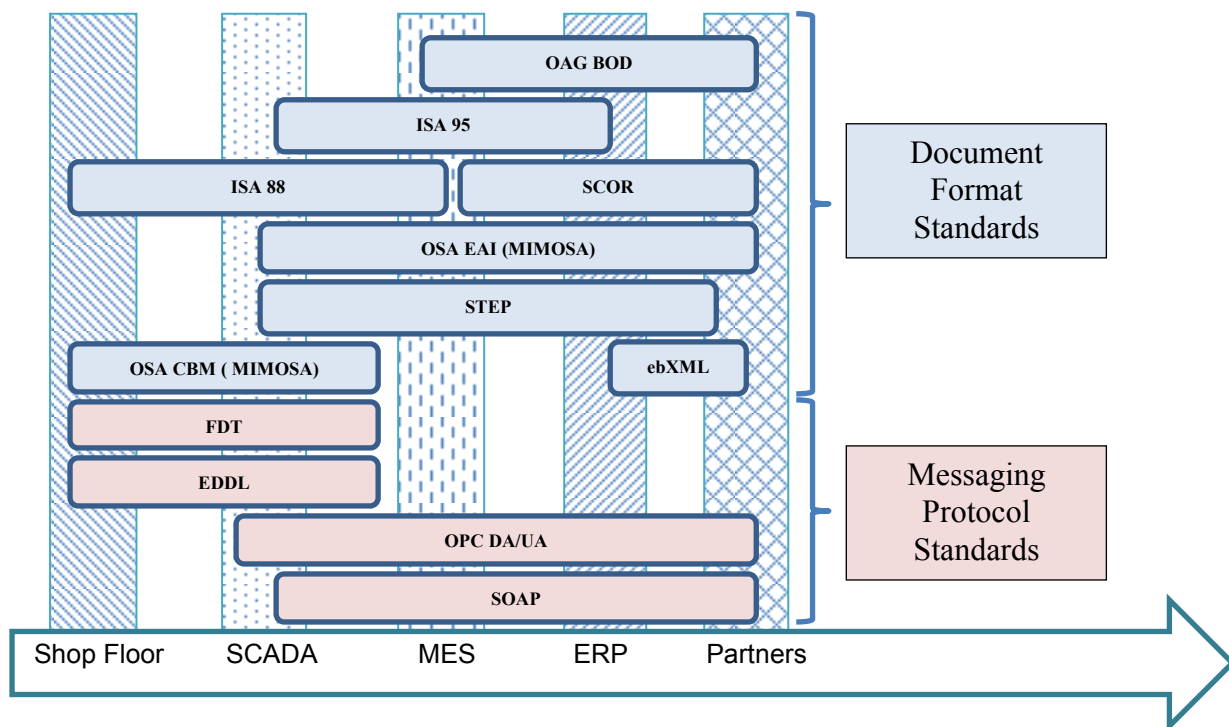


Fig. 1 Manufacturing standards scope diagram

At the technical level, the ESB assures message mediation and data transformation, offering a uniform messaging platform. Most Enterprise Service Bus implementations are providing a business process execution engine that is capable to run BPEL processes. The formalization of business processes enables organizations to move to an orchestrated operational model (Moore et. al 2004).

Another important factor on the market today is the emergence of cloud computing paradigm. The pay-as-you-go model offered by cloud computing is again a game changing factor, allowing a new level of flexibility at the IT layer in support of adopting SOA (Ebberts 2009).

In this paper we will take a look at how these developments are influencing the trends in manufacturing enterprises evolution. We will focus on general aspects and then zoom in to the IBM Manufacturing Integration Framework.

2. MANUFACTURING ALIGNMENT WITH SOA

Almost all industries by now strive to achieve a SOA, either by starting from scratch or by slowly migrating the legacy applications and more importantly legacy processes towards this goal. So, how are the manufacturing enterprises reacting to this trend? First of all the manufacturing enterprises have to move in this direction also. Probably the most important reason is that the market has extended and became global.

This means that a new kind of integration is required with third party partners that are not even known in advance or that are subject to change.

Figure 1 illustrates the most important standards from a manufacturing perspective, together with their scope, starting from shop floor level to the integration with external partners, further divided in document format standards and protocol standards.

2.1 Document Format Standards

OAG-BOD: is a document format standard developed by Open Applications Group Integration Specification (OAGIS) with the goal of encapsulating the data from and to higher level applications on ERP layer and external partners. For example, customer orders are transformed in a standard document (OAG Business Object Document – BOD) and passed to the MES. This allows a decoupling between MES and higher levels, allowing flexibility in regards to the ERP implementation (i.e. ERP system can be replaced with no impact on MES integration). For more information refer to: www.oagi.org.

ISA-95: was developed in order to provide the interface used to exchange data between enterprise systems (ERP, Supply Chain, High Level Scheduling) and manufacturing operations systems (Work Item Scheduling, Plant Data Collection and Analysis, WIP Tracking, Operation Execution). For more information refer to: www.isa-95.com.

ISA-88: was developed by the WBF and is the worldwide recognized standard for batch processing. The standard is implemented by BatchML, which provides a set of XML type and element definitions that can be used for batch, product recipe and equipment data. The latest version of BatchML at this time is V2. For more information refer to: www.isa-88.com.

SCOR: Supply Chain Operations Reference-model has been developed by Supply-Chain Council (SCC) as the cross-industry standard for supply-chain management. The SCOR model scope spreads over all business processes involved in satisfying a customer's request. The five major processes covered are: Plan, Source, Make, Deliver, and Return. For more information refer to: www.supply-chain.org.

OSA-EAI/CBM (MIMOSA): Open System Architecture for Enterprise Application Integration (OSA-EAI) architecture is a specification published by the Machinery Information Management Open Systems Alliance (MIMOSA) organization and is focused on asset management. MIMOSA publishes XML specifications for Enterprise Application Integration (EAI) and Condition-based Maintenance (CBM). For more information refer to: www.mimosa.org.

STEP: Standard for the Exchange of Product Model Data is an ISO standard (ISO 10303) describing exchange of digital product information between computer-aided design (CAD), computer-aided engineering (CAE), and computer-aided manufacturing (CAM) systems. For more information refer to: www.steptools.com.

ebXML: Electronic Business using eXtensible Markup Language set of specifications, that assure a standard method to exchange business messages, conduct trading relationships, communicate data in common terms and define and register business processes. ebXML was started in 1999 as an initiative of OASIS and the United Nations/ECE agency CEFAC. For more information refer to: www.ebxml.org.

2.2 Messaging Protocol Standards

FDT: Field Device Tool standard is needed to ensure the consistent management of a plant-wide control and automation technology, focused on life-cycle management. A goal of the standard is to create universal and central plant-wide tooling for the life-cycle management of heterogeneous fieldbus environments, multi-vendor devices and sub-systems in process and manufacturing. For more information refer to: www.fdtgroup.org.

EDDL: Electronic Device Description Language technology is used by major manufacturers to describe the information that is accessible in digital devices. Electronic device descriptions are available for millions of devices that are currently installed in the process industry. The technology is used by the major process control systems and maintenance tool suppliers to support device diagnostics and calibration. For more information refer to: www.eddl.org.

OPC DA/UA: The OPC-Data Access specification defines a standard set of objects, interfaces and methods to enable process control and manufacturing automation applications to facilitate interoperability, by providing an abstraction layer to hide heterogeneous device protocols.

OPC-Unified Architecture is a relatively new standard proposed by the OPC Foundation, focusing on enterprise integration. OPC-UA is based on SOA, web services, XML based architecture and WS-* standards. The transport protocol is however proprietary, considering performance as a priority (OPC). For more information refer to: www.opcfoundation.org.

SOAP: Simple Object Access Protocol is a protocol specification for exchanging structured information in the implementation of Web Services in computer networks. It relies on Extensible Markup Language (XML) for its message format, and usually relies on other Application

Layer protocols, most notably Hypertext Transfer Protocol (HTTP) and Simple Mail Transfer Protocol (SMTP), for message negotiation and transmission. SOAP can form the foundation layer of a web services protocol stack, providing a basic messaging framework upon which web services can be built. For more information refer to: www.w3.org/TR/soap.

3. IBM PRODUCTS ENABLING SOA/ESB

Helping to integrate open standards environments IBM offers WebSphere ESB, providing an ideal ESB for web services deployment. Key points about WebSphere ESB are:

- It is built on WebSphere Application Server, providing a close affinity for the services hosted within the application server, and able to exploit the robustness, security and scalability of the Application Server.
- It is at the heart of WebSphere Process Server deployment ensuring that the Process Server gets access to the flow of information through the business to be reflected in the business processes.
- It is shipped with WebSphere Adapters, accelerating one's ability to integrate with the business critical applications, delivering more value to one's business faster.

Where there is the need to integrate a broad mix of environments, applications and data formats flowing through many parts of the business, and spanning multiple architectures, IBM offers WebSphere Message Broker, which delivers an Enterprise Service Bus capable of integrating and service-enabling the most heterogeneous IT infrastructures to provide the foundation for a SOA. The four main points regarding this product are:

- The Message Broker's ability to provide universal connectivity by leveraging WebSphere MQ and supporting both standards-based and non-standard APIs and protocols. WebSphere MQ provides reliable, secure, high speed messaging connecting all your assets.
- The ability to offer universal data transformation using a variety of methods, including XSLT, graphical mapping, and leveraging Transformation Extender for complex data transformation tasks.
- New and improved pre-built capabilities that reduce development time, increase user productivity, and hence improve the time to value for Message Broker customers.
- The Message Broker's ability to meet the most advanced performance requirements in terms of transactionality, throughput, and scalability.

IBM's third ESB offering provides an ESB is a different form-delivered as an appliance-WebSphere DataPower XI50 Integration Appliance, the hardware ESB appliance:

- This appliance offers capabilities for services, including great capability for locking stuff down, accelerating key processing, and handling key

integration functions – thus making sure the ESB is really efficient and bullet-proof.

- The appliance - a specialized hardware for the enterprise's SOA - minimizes the complexity and reduces the cycles of implementing the SOA infrastructure, while enhancing performance, security and management of business processes implemented under SOA principles.
- Being able to deploy ESB capabilities as an appliance makes possibly delivering these capabilities anywhere in the network that needs them.

This flexible combination of ESB products, all of which work together across the enterprise helps to deliver an ESB without limits - ensuring that no matter what the in-house skills, the system architecture or the programming choices, the corresponding SOA will be able to benefit from the most complete ESB solution best suited to particular environments and needs. ESB service enrichment results from: matching and routing communications between services; conversion between transport protocols; transformation between data formats; identifying and distributing bus events (Borangiuc 2011).

4. IBM MANUFACTURING INTEGRATION FRAMEWORK (MIF)

While working with manufacturing enterprises from various perspectives over the years, IBM has observed some of the most important challenges they face. As the technology supporting the manufacturing processes has been developed and adopted over the last couple of decades, the current solutions lack in flexibility, by not being able to support process changes/improvements or not being able to offer the data accessibility required. Another important aspect is the fact that some of the technology used in these solutions has become obsolete. Obsolete technology raises risks of production outages as the skills and replacement parts are diminishing over the years. From a cost perspective, the support cost associated with these aging systems can grow exponentially, as it involves proprietary hardware and software, integrated in a point to point architecture. Also, as companies have become larger especially through acquisitions, duplicate functionality exists in different systems in the enterprise.

With these challenges in mind, IBM has developed together with the manufacturing enterprises, initially from automotive domain, the Manufacturing Integration Framework (MIF).

MIF is a solution enablement workbench built on open standards and on SOA technology and it should be understood as a framework rather than a complete application. Various IBM technologies are pre-configured in MIF to provide the building blocks for developing a complete solution to support manufacturing processes. MIF allows integration from plant floor layer and up to ERP business systems and external partners.

4.1 SOA Foundation Enabling MIF

The IBM SOA foundation diagram is illustrated in figure 2.

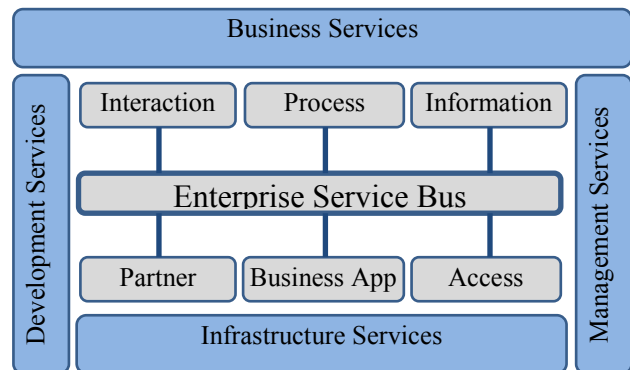


Fig. 2 IBM SOA foundation

This diagram shows how services in a SOA ecosystem can be grouped in various categories. MIF's pre-packaged components are mapped on the SOA foundation as follows:

User interaction services in MIF are provided through a centralized web based portal implementation that allows role based data access. The portal offers dashboard views of the relevant data in each role along with capability to adjust the manufacturing business process configuration at runtime.

Process services are provided by the integrated BPEL engine available with IBM SOA suite. The BPEL engine allows configuring choreographies for manufacturing processes across the shop floor and other enterprise applications and external partners.

Information services are provided by the framework by collecting centrally all manufacturing data and allowing centralized reporting.

Partner services are possible by extending the manufacturing business processes to interact directly with business partners using industry standards.

Business application services are assured by the underlying IBM middleware (WebSphere, Tivoli, Rational, DB2), which assures a robust, scalable and secure services platform.

The access services are provided by the Tivoli Identity and Access Management software suite. These services include role based access control, policy based provisioning of user accounts, attestation and fined grained data access. The access solution enables single-sign-on for both web based and legacy applications used across the enterprise.

The enterprise service bus (ESB) implementation allows message passing (publisher/subscriber), message validation and transformation services and connects software products (from various software vendors), PLCs and MES systems.

Development services include business process declarative development, shop floor event definition and mapping between events and business processes. MIF allows also definition of a dashboard of key manufacturing process

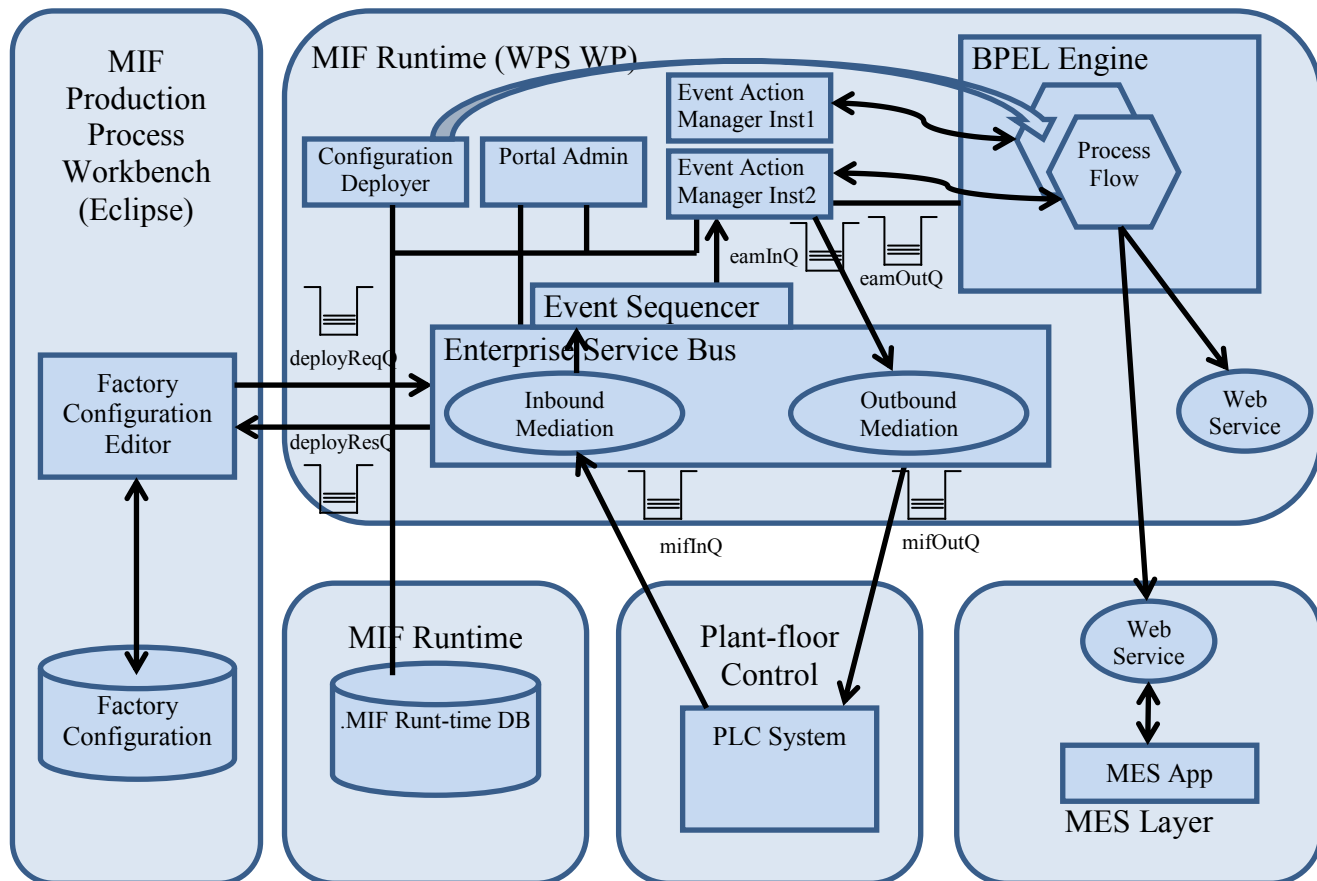


Fig. 3 Manufacturing Integration Framework Architecture

metrics (KPIs) for line and program managers, facilitating business process tracking and continuous improvement.

4.2 MIF Architecture

Figure 3 illustrates the MIF architecture consisting in a workbench application and the actual MIF runtime.

The Production Process Workbench application is based on Eclipse platform and is an integrated development environment that facilitates the MIF implementation. The workbench serves as a design tool, allowing declarative development of manufacturing processes, definition of shop floor events and the binding between these events and the corresponding business processes. Another goal of the workbench is to hide the underlying complexity (BPEL processes, EARs, dependencies) from the person configuring these and basically enables a process control person to implement the factory configuration, with no programming skills or advanced knowledge about the underlying technology. Once the configuration is complete, the factory configuration is stored in the Configuration Database.

The MIF runtime is consisting in the following applications:

Enterprise Service Bus: MIF is using the IBM WebSphere MQ ESB implementation which serves two main roles. The first one is in providing mediation services, which consists in protocol conversion and data transformation. The second role is in assuring temporal decoupling between the PLC systems, configuration database and MIF runtime by message queuing.

The WebSphere MQ binding provided with WebSphere ESB and WebSphere Process Server allows communication between native MQ applications and a Service Component Architecture (SCA) environment. The WebSphere MQ bindings make receiving and sending MQ messages simple, allowing additional manipulation of the message from one application to another.

Event Sequencer: is an extension specially built for MIF. This extension to the ESB allow sequencing of PLC inbound events based on specific rules. The requirement for an event sequencer comes from the fact that normally shop floor events need to be processed in a predefined order. The sequencer acts as a buffer where events can be staged and released in the required order to the Event Action Managers.

Event Action Manager: is responsible with the event wrapping and the invocation of the corresponding BPEL process using the BPEL engine APIs. In order to determine the BPEL process to be invoked, the event action manager checks the event/process bindings in the configuration database.

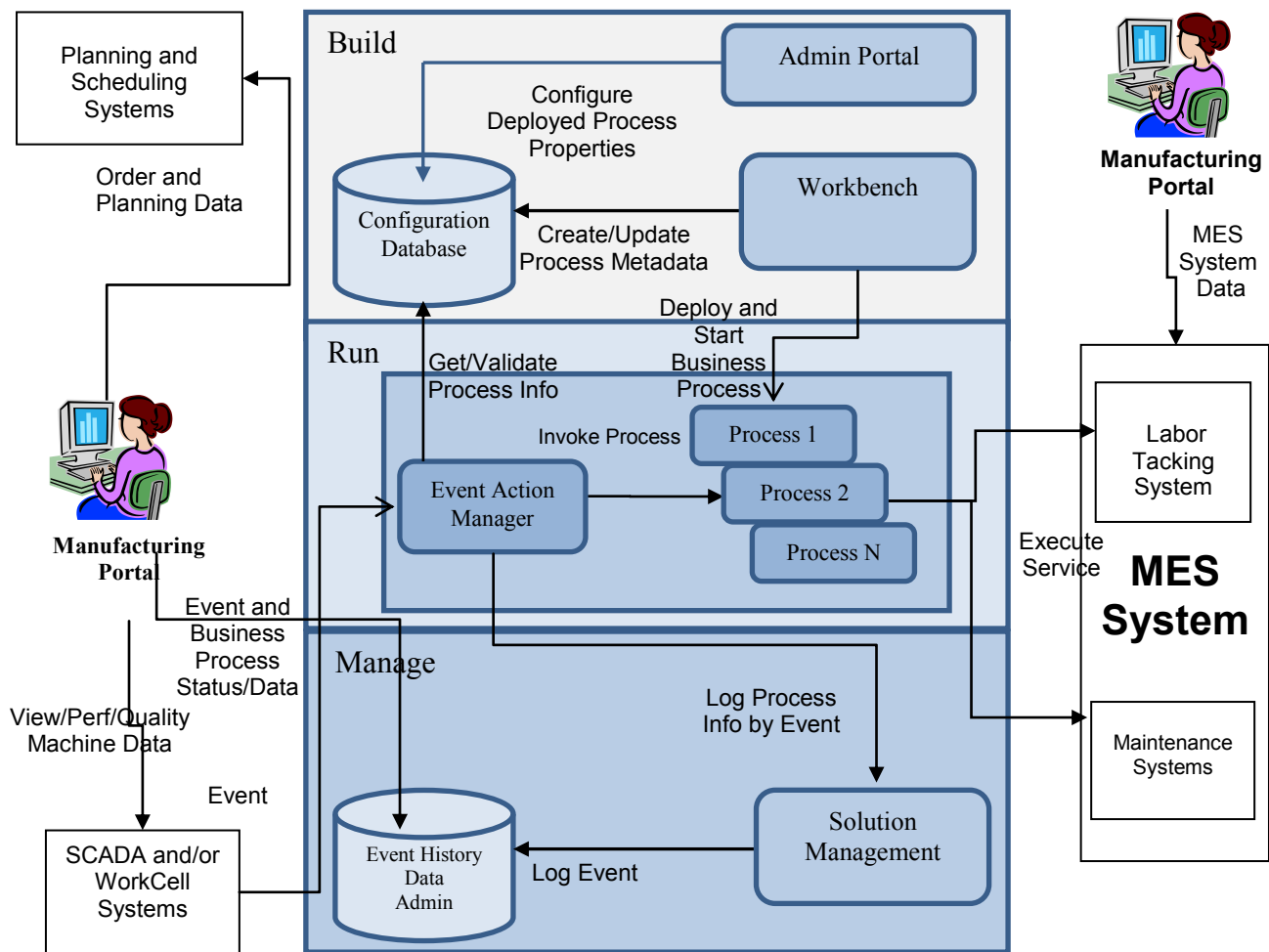


Fig. 4 Manufacturing Integration Framework Layers

BPEL Engine: is the IBM implementation of a BPEL runtime platform. IBM's WebSphere Process Server, or simply named Process Server, includes the business process choreographer, a flow engine based on WS-BPEL. The previous version was called WebSphere Business Integration Server Foundation, which also included WS-BPEL support. Process Server has several major features including:

- **Business processes:** Processes can be stateful and long-running, or transactional micro-flows. Long-running processes can't be rolled back like micro-flows, however they can have compensation handlers to undo previously performed activities. Processes can be used to implement composite services.
- **Human tasks:** A key part of a business process is the ability to bring people into the process. The human task manager enables steps with people to be invoked as a service. Workflow patterns are supported through either external (participating) tasks, or inline tasks using a BPEL extension.

- **Business Rules:** The integrated rules engine enables business rules to be created and evaluated, rather than hard-coding the decisions into the business process. Authorized users can update the rules using a Web browser. Administrators can activate the updates, and export them so the development environment can be kept in sync with the run-time.
- **SCA:** Service invocations in Process Server are done using the Service Component Architecture (SCA) specification. SCA interface maps can be used to invoke services with different interfaces than the calling component.

4.3 MIF Runtime Dataflow

The manufacturing enterprise service bus (ESB) is made up of three main layers, as illustrated in Figure 4:

The Build layer involves the component called the 'Production Process Workbench' (PPW). This is an Eclipse based tool which allows the user to define a manufacturing line and assign business processes to points on the line. The PPW allows users (of different roles) to do the following:

- Create a tree view of manufacturing enterprise

- Create service categories and import pre-existing services into those categories
- Create event input and output messages
- Create business processes from imported services – these processes become an ‘eventName’
- Associate defined processes/events to points on the manufacturing enterprise tree (these points become an ‘eventSource’)
- Deploy newly created or updated business processes to the runtime engine dynamically

The Run layer involves the component called the ‘Event Action Manager’ (EAM). This is a set of EJBs running in the application server which are responsible for invoking the correct business process when an event comes into the bus. The EAM does the following for each event that arrives:

- Accepts either JMS or web service type events
- Allows for one interface into any business process, rather than a specific interface for every process created in the bus
- Substitutes default parameters into the event message for overridden or missing parameters
- Logs event details using Common Event Infrastructure paradigm

The Manage layer involves the component called the ‘Solution Manager’. This consists in a set of portlets and EJBs which are responsible for getting/viewing event information. The Solution Manager has the following capabilities:

- Captures event/process data in a standard format for display
- Displays event details with searchable portlets
- Facilitates the recovering or restarting of failed events

The MIF implementation is an on-going process as the main point of MIF is to assure flexibility considering that the manufacturing requirements are changing continuously. The MIF implementation process starts in the workbench where the configuration manager models the whole manufacturing process, all the workstations, the process points in manufacturing, which events will occur in those process points and as results of these events what business process will be executed and the deviations that need to occur in order to transform the data.

Once the initial configuration is completed by the configuration manager, it is then deployed to the runtime server where the configuration deployer creates the runtime artefacts that are necessary to support that specific configuration and that will be generating the runtime BPEL EAR files. Then the EAR files will be deployed and started on the Process Server and will update the database with the current configuration. At that point the portal admin functionality allows the user to customize the configuration,

to set input and output default values and schedules. Also it permits enabling and disabling events on the fly.

Once the business processes are deployed and started, the new configuration is ready to start receiving incoming events from the plant floor, from the PLC or from another application. When the events come in they are accepted by an Inbound Mediation component. The Inbound Mediation recognizes the incoming event and transforms it into the correct format for the workflow. The event is then released to the Even Action Manager. The Event Action Manager wraps the event and invokes the BPEL runtime engine. All the events are register with their initial and final state, the result, in the database which allows to see the health of the system or to resubmit failed events.

From standards perspective, MIF maps to ISA 95 standard. At a higher layer MIF leverages OAG BOD. The integration with shop floor devices is governed by ISA 88 and OSA CBM (MIMOSA).

5. CONCLUSIONS

MIF is the first formalized framework for implementing SOA in manufacturing enterprises. MIF is build based on years of experience gathered by IBM from working closely with leading manufacturing enterprises in automotive domain like Ford, Honda, Volkswagen and many others. In the last years, MIF is being implemented in other domains like aerospace or electronics.

The main characteristic of implementing MIF is the fact that is a continuous process that starts with the initial implementation of ESB layer and BPEL processes, and continues with the implementation of the changes required to adapt to the new requirements dictated by the business drivers. The framework provides this ability to react and implement changes with low or even zero programming effort and it does this by providing tools which allow declarative development production lines and of the business processes associated with shop floor events.

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